

2.3 Machine Learning vs. Deep Learning

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Introduction

Machine learning and deep learning are both subfields of artificial intelligence in which algorithms learn patterns from data. They differ in how they extract features, how much data and computing power they require, and the kinds of problems they are best suited to. In traditional machine learning, people manually select the features that matter and pass them to algorithms such as logistic regression or decision trees. In deep learning, networks with many layers discover features automatically from raw input, without that manual step (Indiana Wesleyan University, 2024).

This report examines two examples drawn from the course lesson — email spam detection as the traditional machine learning example and image recognition as the deep learning example. For each, I identify a real-world application, explain why the chosen approach is suitable, and explain why the alternative approach would be less suitable.

Case Study 1: Email Spam Detection (Traditional Machine Learning — Logistic Regression)

Problem Description

Email spam detection is a binary classification problem: for each incoming message, a system must predict whether it is spam or not spam. The model learns from labeled examples and uses that learning to classify new messages. Because the two outcomes are mutually exclusive and the features that distinguish spam from legitimate email can be identified in advance, this type of problem is well matched to traditional supervised classification algorithms.

Selected Traditional ML Approach: Logistic Regression

The course lesson identifies email spam detection as a traditional machine learning example and lists logistic regression (or decision tree) as a suitable algorithm for this type of problem (Indiana Wesleyan University, 2024). Logistic regression estimates the probability that an input belongs to one class — in this case, the probability that a message is spam. Engineers define a set of features in advance: the frequency of certain words, the presence of links, the sender domain,

message length, and similar measurable characteristics. The model learns which combination of those features is most predictive of spam during training.

Real-World Application: Gmail

Gmail, Google's email service, uses machine learning to filter spam and protect users from phishing. Email spam detection is therefore a documented real-world application of machine learning for binary classification. Gmail demonstrates the real-world use of machine learning for spam detection. Logistic regression is evaluated here as a suitable traditional classification approach based on the course lesson; this report does not claim that Gmail currently uses logistic regression in its production system.

Why Logistic Regression Is Suitable

Binary classification with structured inputs. Spam detection has exactly two outcomes. Email messages can be converted into structured numerical inputs — word frequencies, presence of links, sender characteristics — that fit logistic regression's requirements directly. The course lesson describes this feature-selection approach as characteristic of traditional machine learning (Indiana Wesleyan University, 2024).

Class probabilities. Logistic regression outputs a probability score, not just a hard label. That score can be used to tune the spam-detection threshold, making it easier to balance false positives against false negatives for a given use case.

Relative interpretability. Each feature in a logistic regression model has an associated coefficient that indicates its contribution to the prediction. This makes it possible to examine why particular messages are flagged, which supports auditing and adjustment over time. Interpretability is not unlimited — it still depends on how the features are constructed — but logistic regression is more transparent than many other models.

Practical efficiency. When the important features are already known and can be engineered before training, logistic regression is efficient and computationally inexpensive compared to deep learning. The course lesson notes that traditional machine learning "can work effectively with smaller datasets and less processing power" (Indiana Wesleyan University, 2024). For a structured classification problem where the feature representation is already available, this efficiency is an advantage.

Why Deep Learning Would Be Less Suitable Here

Deep learning could technically be applied to email spam detection, but it would be less suitable for this type of problem. Its main strength is automatic feature extraction from raw, unstructured data — a strength that is not needed when the relevant features can already be defined and structured manually. Applying a multi-layer neural network here would increase model complexity, require more training data and computing resources, and produce a model that is harder to interpret. The performance improvement, if any, may not justify the additional implementation and maintenance cost for a binary classification problem that simpler methods handle well. The lesson captures this directly: deep learning becomes valuable when the task is too complex for a checklist approach, not when the checklist already works (Indiana Wesleyan University, 2024).

Case Study 2: Image Recognition (Deep Learning — Convolutional Neural Network)

Problem Description

Image recognition is the task of identifying objects, people, or scenes in a photograph or video frame. Unlike spam detection, where features can be defined and measured in advance, an image arrives as a grid of raw pixel values. Those pixels carry no pre-attached labels or feature names. The system must learn, entirely from examples, how to turn those numbers into meaningful visual output — recognizing a face, an object, or a location.

Selected Deep-Learning Approach: Convolutional Neural Network (CNN)

The course lesson identifies image recognition as a deep learning example and specifies convolutional neural networks as the algorithm: "In facial recognition systems, a CNN learns to detect and recognize faces from raw pixel data without needing predefined features like eye or nose shapes" (Indiana Wesleyan University, 2024). A CNN processes an image through multiple layers. Earlier layers detect low-level patterns such as edges and color gradients. Middle layers combine those into shapes and textures. Deeper layers recognize whole objects or faces. This automatic, hierarchical feature extraction is what makes CNNs effective for visual data.

Real-World Application: Google Photos

Google Photos uses AI to group similar photos together, organize a library by people, pets, places, and objects, and enable users to search their photo collection using recognized content. Google has publicly described AI-powered face grouping and photo organization as confirmed product features (Google AI, n.d.; Krovblit, 2023). These capabilities are documented real-world applications of deep learning for image recognition.

Official Google sources document AI-supported photo organization and face grouping in Google Photos. CNN research is used here to explain why deep learning is suitable for image-recognition problems; it is not presented as confirmation of Google Photos' exact internal architecture. Research on CNN-based face recognition — such as FaceNet, developed by Schroff, Kalenichenko, and Philbin (2015) at Google — demonstrates the kind of technical approach that makes this class of product feature possible. This report uses that research to explain CNN suitability, not as a claim about any specific model running inside Google Photos.

Why Deep Learning Is Suitable Here

Images are raw, high-dimensional, unstructured data. A photograph contains millions of pixel values with no inherent labels or structure. No practical set of hand-crafted rules can reliably describe what a face, a dog, or a mountain looks like across different lighting conditions, angles, scales, and backgrounds. The lesson notes that in deep learning, "the algorithm learns the features by itself from raw data" — which is necessary when those features cannot be pre-specified (Indiana Wesleyan University, 2024).

Hierarchical feature learning. Recognizing a face requires understanding patterns at multiple levels of abstraction — from pixel edges to contours to facial geometry to identity. CNNs are designed to build this kind of layered representation automatically. Schroff et al. (2015) demonstrated that a CNN trained to map face images to compact numerical embeddings can then support recognition, verification, and clustering tasks with high accuracy. Traditional ML classifiers have no equivalent mechanism for learning features in stages.

Performance under real-world variation. Real photos vary in lighting, camera type, image resolution, and subject angle. CNNs trained on diverse image data can learn features that remain

effective across this variation. Achieving similar robustness with hand-designed features would be extremely difficult (Schroff et al., 2015).

Why Traditional ML Would Be Less Suitable Here

Traditional machine learning could be applied to image recognition, but it would be less suitable for this type of problem. A traditional classifier requires features to be extracted from each image before training — for example, by measuring estimated distances between facial landmarks, computing color histograms, or detecting predefined edge patterns. The lesson identifies this manual feature selection as the core characteristic of the traditional ML approach (Indiana Wesleyan University, 2024).

For real-world photographs taken under varied, uncontrolled conditions, hand-designed features would be difficult to define accurately across all the relevant variation in lighting, angle, scale, background, and resolution. Features that work well in one setting may fail in another.

Traditional classifiers also lack the architecture to build visual representations in layers — they can only score the features they are given. A separate feature-engineering pipeline would need to be designed, maintained, and updated as the range of images changes. These constraints make traditional ML less practical than deep learning for image recognition at useful accuracy levels, even if it is not technically impossible.

Comparing the Two Approaches

The two case studies illustrate that the right choice between traditional machine learning and deep learning depends on the structure of the data and the problem. When the important features are already understood and can be converted into structured inputs — as in spam detection — traditional ML methods are efficient, interpretable, and well-suited to the task. When the input is raw and unstructured and features must be discovered rather than defined — as in image recognition — deep learning's automatic feature extraction is what makes the problem tractable.

The course lesson expresses this contrast directly: traditional ML "is like giving a student a checklist of things to look for in an email to decide if it's spam," while deep learning "is like teaching the student how to read and letting them figure out what makes an email spam based on experience" (Indiana Wesleyan University, 2024). Neither approach is universally superior. Each works best in a different kind of situation.

Conclusion

Email spam detection and image recognition are useful examples for understanding when traditional machine learning and deep learning are each appropriate. Logistic regression is well-suited to spam detection because the relevant features are structured and can be defined in advance, the problem requires binary classification, and the model's outputs and coefficients are relatively transparent. CNNs are well-suited to image recognition because useful visual features cannot be manually specified from raw pixel data and must instead be learned from examples. Knowing which approach fits a given problem is one of the most practical concepts in applied machine learning.

References

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